Human Activity Recognition on Smartphone

Technical Assessment — Reev | Van Quang LUU | 10/06/2026

1. Dataset Description

I collected IMU data using Phyphox (RWTH Aachen) on my Samsung smartphone, placed in the front pants pocket. Accelerometer and gyroscope were recorded separately at ~100 Hz and merged onto a shared uniform time grid. Walking and running were recorded outside on the street at normal pace / slow jog. Sitting sessions were split between desk and couch. Falls were simulated as controlled forward and lateral drops on a mattress.

Activity	Sessions	Details
Walking	3	~4 min, street, normal pace
Sitting	3	~4 min, desk (×2) + couch (×1)
Running	3	~4 min, street, slow jog
Falling	1	~1 min, forward + lateral on mattress

2. Preprocessing & Feature Engineering

Resampling: Phyphox timestamps are irregular and the two sensors don't share the same grid, so both are linearly interpolated onto a uniform 100 Hz grid. **Noise filtering:** A zero-phase Butterworth low-pass filter (order 4, cutoff 20 Hz, via *filtfilt*) is applied to each channel. Human motion energy lies below 15–20 Hz, so this removes sensor noise without distorting the movement signal. **Windowing:** 2-second windows with 50% overlap (hop = 1 s). Two seconds captures 2–3 full gait cycles; a fall fits in one window (~1 s). The overlap doubles the number of examples.

Features: 105 features per window — 12 time-domain stats (mean, std, RMS, energy, kurtosis, ZCR…) and 3 frequency-domain stats (dominant frequency, spectral energy, spectral entropy) for each of the 6 channels, plus the same 15 on the acceleration magnitude (orientation-invariant norm of acc x/y/z).

Split strategy: Session-level split, not window-level. Adjacent windows share 100 samples (50% overlap), so a random window split would leak near-duplicate data across the boundary. Sessions are assigned whole: last session → test, second-to-last → val, rest → train. For falling (single session), a contiguous 60/20/20 block split is used. Final counts: **train 328 • val 198 • test 198** windows.

3. Model Architecture & Training

Random Forest was chosen over deep learning (LSTM, 1D-CNN) because the dataset is small (~700 windows), no feature scaling is needed, inference is very fast, and the serialized model stays lightweight. class_weight='balanced' handles the under-represented falling class without manual oversampling. The validation set guided hyperparameter tuning; the test set was evaluated once at the very end.

Hyperparameter	Value	Justification
n_estimators	100	Stable predictions, lightweight
max_depth	12	Limits overfitting on small dataset
min_samples_leaf	3	Avoids memorising noisy windows
class_weight	balanced	Compensates for under-represented falling class

The model takes 105 features as input (6 channels × 15 features + 15 on acceleration magnitude) and random_state=42 ensures full reproducibility.

4. Results

Class	Precision	Recall	F1	Support
Walking	0.98	0.98	0.98	64
Sitting	1.00	1.00	1.00	60
Running	1.00	0.98	0.99	65
Falling	0.90	1.00	0.95	9
Macro avg	0.97	0.99	0.98	198

Test accuracy: 99% — Macro F1: 0.981

Failure modes: Sitting is perfect (F1=1.00) — signal is nearly flat. Walking/running show 1–2 confused windows, likely at transition pace (slow jog ≈ fast walk). Falling has recall=1.00 (no fall missed), which is the critical metric in a medical context. Precision=0.90 means one false alarm, but with only 9 test windows a single error shifts precision by ~10 pts — more falling data is needed to draw firm conclusions.

5. On-Device Performance

Metric	Value
ONNX model size	40.4 KB
Mean inference latency	0.032 ms / window
Real-time budget (window hop)	1000 ms
Real-time headroom	31 354×

With a 1 s hop between windows, the model at 0.032 ms is ~31 000× below the real-time budget — comfortable even on a much weaker embedded CPU. ONNX and sklearn predictions match exactly (agreement = 1.0000).

6. Limitations & Next Steps

All data comes from one person (me), which likely explains the high accuracy but limits generalization — a real deployment needs data from multiple subjects with different gaits. The falling class is also limited to controlled mattress falls, which may not reflect real-world dynamics. With more time I would collect multi-subject data, add signal augmentation (jittering, time-warping), and compare with a small 1D-CNN.